Phase I Autonomous System Decision Logic

Autonomous development was executed as an auton-only effort beginning on 10/05 using a drive-base-only prototype. Due to mechanical limitations of the initial robot configuration, the strategy prioritized **robustness and repeatability** over routine complexity to support state qualification based upon the robotic capacity. From October through early January (01/23), development focused on **iterative validation under Sate Level Qualify competition conditions**, control simplification, and optimization of timing-based execution to ensure consistent autonomous performance.

Design Summary – Phase I

Phase I Autonomous System Time-Line

10/05 – Autonomous development initiated using **drive-base–only** motion to establish a stable baseline for repeatability.

Oct–Nov (First competition: Nov 22) – With a single intake and no match loader, a distance-based autonomous was implemented. Testing revealed performance degradation due to heading drift and intake inconsistency under match conditions (Skills Auton: 2 AP / 26 pts).

12/10 – Following a gear ratio change and intake timing redesign, autonomous logic was updated to match the new robot configuration. Acceleration control and intake-tube timing were explored. Match loading became possible but remained insufficiently reliable, and limitations were formally documented (Skills Auton: 0 AP / 26 pts).

Late Dec–Early Jan – A strategic shift was made from distance-based to time-based autonomous control in response to motor response variability, field surface friction, and cumulative heading drift, resulting in significantly improved consistency and repeatability (Skills Auton: 26 AP / 65 pts).

By 01/24 – Autonomous routines were restructured to separate left- and right-side execution paths. The resulting code achieved competition-level reliability consistent with state qualification performance (Skills Auton: 28 AP / 55 pts). Variability caused by manual preload placement was identified as a dominant error source and aiming error with higher speed, leading to the decision to introduce sensor-based autonomous control in increase the tolrance in Phase II.

Design Summary – Phase I

Phase I Autonomous System Diagram

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Design Summary – Phase I

Phase I Autonomous Lead System
- Block Debug Flow Chart

Auton fails?
↓
Does robot place expected intial location ?
├─ NO → Adjust location (add in check list)
└─ YES
↓
Is robot shaky?
├─ Yes→ Reduce acceleration/ speed
└─ No
↓
Does robot move expected distance?
├─ NO → Adjust time duration
└─ YES
↓
Is heading consistent?
├─ NO → simplify turn/ adjust and angle
└─ YES
↓
Is intake (including march loader) consistent?
├─ NO → Adjust intake timing window
└─ YES
↓
Is outtake consistent?
├─ NO → Adjust outtake timing window
└─ YES
↓
Run full auton validation

Design Summary – Phase I

Phase I Autonomous System overview

Category	Design	Known Limitations	Future Improvement
Distance based	Original auton code with robot able to go in all directions and intake/outtake motors	Not always accurate and consistent with Mechanical limitation (ball jamming)	Make robot more reliable and increase capacity in general
Distance based	Acceleration code for auton	Can't work because of drift needed this code. However, the code wasn't right because the distances are different	Debug acceleration code
Time based	Driving code is time-based instead of distance-based	Need to be revise on how much time to give the robot	Include sensors Skill auton reliabe in 30 s can increase more action to rest 60s